

Evaluation of the LDACS Cybersecurity Implementation

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Abstract—Communication, Navigation and Surveillance (CNS) infrastructure in civil aviation must evolve as fast as possible to cope with all challenges posed by the growth of the worldwide population, globalization and the worldwide demand for more and more mobility. Analogue systems are replaced by digital means and automation is becoming much more important to be able to handle the large amount of new entrants in the air traffic system. Spectrum saturation must be solved by introducing digital systems and safety and security of the safety critical infrastructure surrounding civil aviation must be constantly updated to support the ever-growing system complexity. As one of the Future Communication Infrastructure (FCI) candidates we introduce LDACS as the very first true integrated CNS system worldwide. In previous works we have already analyzed its cybersecurity, developed a cybersecurity architecture with corresponding algorithms and demonstrated the improvement in cybersecurity due to our security additions. Here we implement the LDACS cybersecurity architecture and evaluate the impact of introduced security overhead on the LDACS system. We conclude that the proposed protection mechanisms successfully mitigate previously identified risks and only add minor time, data and computation overhead on top of the LDACS protocol stack. With this result, the security solutions are a good candidate to be included in the SESAR wave 2 updated LDACS specification.

Index Terms—LDACS, Cybersecurity, FCI, Digital Datalink, Security Architecture

I. INTRODUCTION

In its latest Challenges of Growth study EUROCONTROL estimates that civil air traffic will grow by a significant rate of 84% until 2040. Under this estimation it is expected that legacy Air Traffic Management (ATM) systems will surpass their capacity limits [7].

Saturation of the VHF band, spectrum depletion, new entrants such as drones in the controlled air space and the growth of the entire worldwide population with increasing demands for transportation, pose difficult challenges to the existing ATM system. New technologies enabling automatized transportation of goods or people to be able to cope with the growth [7], [32], [34] must be developed. One answer to enable quicker, faster and safer handling of air traffic participants is to successfully conduct the transition from analogue to digital communication in worldwide ATM for Communication Navigation and Surveillance (CNS) systems [33].

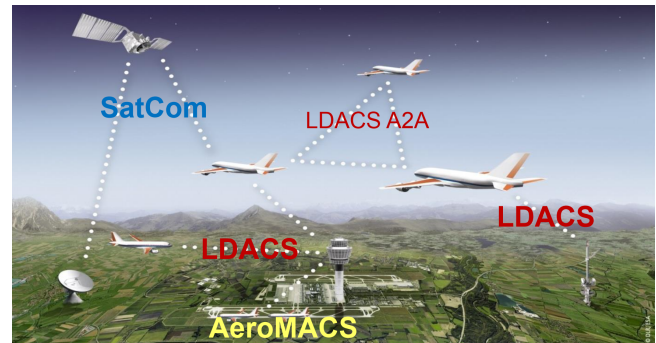


Fig. 1. Future Communications Infrastructure (FCI) [27]

With the Single European Sky ATM Research (SESAR) program of the EU and its counterpart NextGEN in the US, several new digital aeronautical communication technologies are currently under development [7], [34]. The main goal of both research programs is to develop enabling technologies to cope with the expected air traffic growth. These technologies shall make more efficient use of the limited aeronautical spectrum by transitioning from analogue voice to digital data communication [33]. Candidates for the Future Aeronautical Communication Infrastructure (FCI) are AeroMACS for airport communications, SatCOM for oceanic, polar and remote domains, and LDACS (L-band Digital Aeronautical Communications System) for long-range terrestrial aeronautical communications system. Figure 1 illustrates where these candidates are envisioned to be located. LDACS is developed in Europe and currently under standardization by the International Civil Aviation Organization (ICAO). LDACS, as terrestrial digital wireless communication system, enables communication, navigation and surveillance at the same time and by that it is the world's first true integrated CNS system [27]. The communication capabilities of LDACS consist of two parts: (1) the Air-Ground (A/G) link for long-range terrestrial communication and (2) a link for Air-Air (A/A) communications currently developed by DLR.

In this paper we focus on the A/G link. It ensures safety and regularity of flight and is based on technologies used in the

4G mobile phone network [30]. It has been adapted for safety critical infrastructure requirements. With the augmentation of analogue systems by digital substitutes and the related trend towards an increased autonomous data processing as justified in [33], LDACS requires a thorough cybersecurity analysis and cybersecurity architectural design [18]. The envisioned analysis and design should be similar to the ones used in 4G or AeroMACS [4], [10].

In [1], [18], [20] a preliminary draft of a cybersecurity architecture for the LDACS A/G link was justified and presented. In these references it was clearly stated that essential requirements were imposed by the aeronautical spectrum environment (e.g., reduced spectrum bandwidth and limited data rates). In order to develop a successful cybersecurity solution for LDACS, existing security protocols and algorithms had to be analyzed in terms of resource consumption and compatibility. Thus, this paper presents a full software implementation of the LDACS protocol, including its security additions. These simulations were done in order to estimate the additional overhead security brings to the current LDACS protocol. Furthermore we present an extensive overview of LDACS capabilities and ongoing works on LDACS

This paper is structured as follows: Section II presents an extensive overview of LDACS capabilities and ongoing works on LDACS, before the new LDACS's cybersecurity architecture is introduced in Section III. Implementation details with focus on software simulations of LDACS are presented in Section IV. The gained results are explained in Section V before concluding the paper in Section VI.

II. BACKGROUND ON LDACS

The L-band Digital Aeronautical Communications System (LDACS) is the terrestrial data link in the Future Communications Infrastructure (FCI). In 2007 the Federal Aviation Administration (FAA) and EUROCONTROL started a joint investigation on existing technologies to investigate if any technology could fulfill the demands of growing ATM in the future. In short, the answer was no and, thus, sparked the development of LDACS [6].

A. System Characteristics

The realized LDACS has its origin in merging parts of the B-VHF [3], B-AMC [24], [28], TIA-902 (P34) [15], and WiMAX IEEE 802.16e technologies [5]. In 2007 the spectrum for LDACS was allocated at the World Radio Conference (WRC), which is shown in picture 2. It was decided to allocate the spectrum next to DME, resulting in an inlay approach between the DME channels for LDACS as illustrated in Figure 3. Furthermore, LDACS uses LTE/4G technology (e.g., Orthogonal Frequency-Division Multiplexing) (OFDM) to remain highly flexible and scalable and efficient in coding, supporting adaptive coding and modulation. Additionally, it applies (Frequency Division Duplexing (FDD), because of the limited bandwidth available with the inlay approach. Besides all these, LDACS supports seamless handovers, data and voice

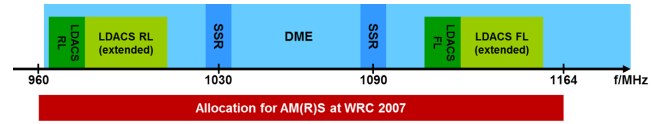


Fig. 2. Frequency assignment for LDACS at WRC 2007, next to DME

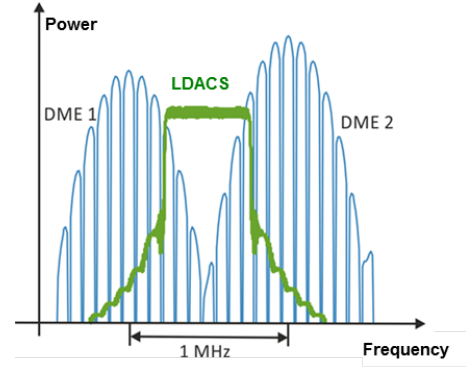


Fig. 3. Inlay approach for LDACS in between the DME bursts

transmissions, Quality-of-Service (QoS), has a navigation and surveillance extension, and an air-air (A/A) link is currently being developed. Most of those points will be clarified in the next subsections.

B. Communication Functionality

LDACS was especially designed for Air Traffic Control (ATC) and ATM applications like Controller Pilot Data Link Communications (CPDLC), Automatic Dependent Surveillance - Contract (ADS-C), full 4D trajectories exchange and real-time weather information such as the Graphical Weather Service (WXGRAPH). It is envisioned that Ground Based Augmentation System (GBAS) functionality will be also be provided via LDACS in the future as well [8]. The underlying enabler for all those applications are the main LDACS parameters listed in Table I. Thus, LDACS covers current Air Traffic Services (ATS), Aeronautical Operational Control (AOC) data and also future applications, enables new concepts (e.g., sectorless ATM) and has at least 50 times more net capacity than VHF Data Link (VDL) Mode 2 [8].

The latest extension of LDACS communications capabilities – the A/A communication – is researched in the German national project IntAirNet [27]. The goal is to establish direct A/A communications between aircraft in communication range allowing infrastructure-less aeronautical network for ad-hoc

TABLE I
MAIN PARAMETERS FOR LDACS

Number of sub carriers	64 (50 used)
Bandwidth	625 / 488 kHz
Subcarrier spacing	9.765625 kHz
OFDM symbol duration	102.4 μ
Guard interval	(4.8 + 12.8) μ
Net data rate	550 kbits - 2.6 Mbit/s

networks between aircraft. However, as already mentioned, this research is only at an early stage, thus we will focus solely on the LDACS A/G link.

C. Entities

A basic LDACS network is formed by multiple Aircraft Station (AS), which are served by a Ground Station (GS) and multiple GS, in turn, are connected to a Ground Station Controller (GSC). All of this is also depicted in figure 4.

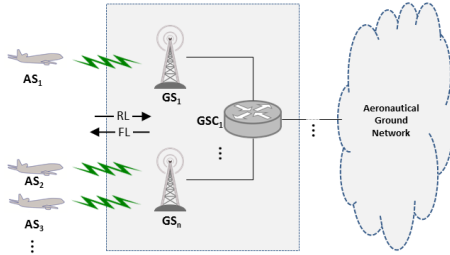


Fig. 4. LDACS network with AS, GS and GSC and the supporting aeronautical ground network infrastructure.

As mentioned in [26] 512 aircraft can be served by one GS where the GS sends a continuous data stream in the Forward Link (FL) to the AS. The Reverse Link (RL) consists of individual bursts of data from each AS to GS. This means, for every RL communication the AS first needs to request the respective resource allocation within its cell from the GS before being able to send. Both FL and RL communication, including user and control data, is done via the air gap over the radio link between AS and GS. On the ground a GSC is responsible for serving several GSs, forming an LDACS sub-network with its LDACS internal ground network infrastructure. The GSC is linked via several intermediary networks, which are out of scope for this paper, finally to the aeronautical ground network. The aeronautical ground network (previously the Aeronautical Telecommunications Network (ATN)) is used for example by Air traffic Network Services Providers (ANSP) and airlines to exchange Air Traffic Service (ATS) or Air-line Operational Control (AOC) data between the ground infrastructure and the aircraft.

D. Protocol Stack

Figure 5 shows the protocol stack of LDACS as implemented in the AS and GS. It consists of the Physical Layer (PHY), the Medium Access Layer (MAC), Voice Interface (VI), Data Link Service (DLS) layer, LDACS Management Entity (LME) and Sub-Network Protocol (SNP) layer. For more details see Section III. The LDACS protocol stack is located on Layer 1, 2 and 3 of the ISO/OSI model with TCP/IP and IPv6 placed above the SNP. Below the SNP, data is fragmented, prioritized and aggregated at the DLS layer. In order to communicate, LDACS uses several logical channels in the MAC layer [26]:

- The GS announces its existence and several necessary physical parameters in the Broadcast Channel (BCCH) to incoming AS.

- The Random Access Channel (RACH) enables the AS to request access to an LDACS cell.
- In the Forward Link (FL) the Common Control Channel (CCCH) is used by the GS to distribute and grant access to system resources.
- The reverse direction is covered by the Reverse Link (RL), where aircraft need to request resources (in so called resource allocation) in order to be allowed to send any data. This happens via the Dedicated Common Control Channel (DCCH).
- User data itself is communicated in the Data Channel (DCH) on the FL and RL.

E. Extension Towards Navigation and Surveillance

Around 2011 the development of an extension towards navigation in the sense of "Alternative Positioning Navigation and Time" (APNT) started [29]. The concept was proven in theory in 2012 with simulations in 2013 showing achievable accuracy of the aircraft position of around 4m [21]. Later in 2013 and 2015, flight trials confirmed the localization accuracy of around 15m with demonstration equipment, while in 2019 it was set out in the German national project MICONAV [14] to test real-time positioning. All of the aforementioned steps confirm, that LDACS can be used as a true navigation substitute for En-Route (ENR) continental flights in case of DME or GPS outage. In the long term, it must be considered whether LDACS can truly replace legacy DME systems in the long-term. There are also considerations to use LDACS for surveillance such as concepts by Filip et al. [9], showing the use of LDACS as a passive radar substitute. However, the easier approach to implement surveillance into LDACS is to use ADS-C and Automatic Dependent Surveillance - Broadcast (ADS-B) and broadcast the position of an aircraft to all neighboring planes and periodically send the position of the aircraft down to the ground network.

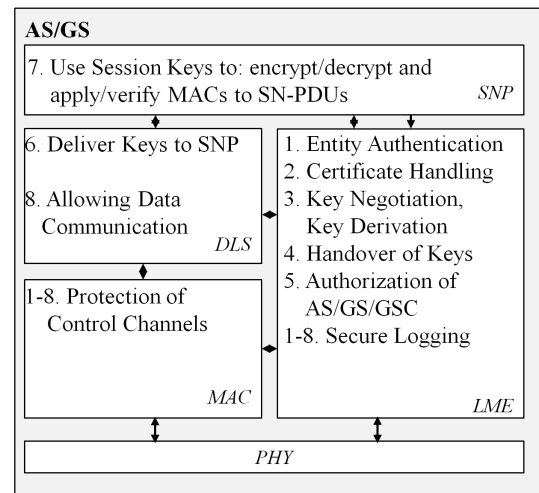


Fig. 5. Protocol Stack of LDACS

F. Summary of key features of LDACS

As described throughout Section II the following steps towards LDACS were designed, implemented, and performed:

- The system design and optimization via simulations including interference mitigation were the first step.
- The LDACS protocol design, starting from layer two in the ISO/OSI stack, is done.
- Navigation functionalities of LDACS were included and proofed and even flight-trialed.
- The LDACS system specification under the lead of EUROCONTROL and SESAR has been completed.
- From there, the technology was transferred to the industry (Frequentis, Rhode & Schwarz and LEONARDO).
- And lastly DLR performed an evaluation of the compatibility with other L-band systems.

As test were successful standardization of LDACS itself started in 2016 with ICAO in the working group "ANB/AN: DCIWG PT-Terrestrial" [17]. As an important step, the LDACS Standards and Recommended Practices (SARPS) were agreed upon in October 2018. The plan for the future is to release guidance material by the end of 2022 in order to have LDACS applied 2024. Then LDACS will be the world's first realized true integrated CNS system.

III. LDACS'S CYBERSECURITY ARCHITECTURE

Due to the fact that digitalization rises also in importance in aeronautical communication [1], [31], [35], security threats increase in parallel as pointed out in safety and security analyses [11], [18]. These results forced DLR to address this security issue for LDACS as soon as possible by drafting a cybersecurity architecture for LDACS as detailed described in [20]. As presented in [19] the SESAR 2020 Security Analysis (SecRAM) process in 2018/2019 revealed new findings on security threats (e.g., DoS on PHY layer, malicious operation software) and recommended a first set of algorithms overcoming them and improving LDACS. These recommendations were integrated into LDACS's cybersecurity architecture as described in detail in [19].

A. Security Features and Message Flow

Figure 5 illustrates the LDACS stack and as it can be seen in each stack component a feature for the realized cybersecurity architecture is included. As we already published two extensive works on the key security functionalities envisioned for LDACS [19], [20], we only give a brief overview of each here.

1) Protection of Broadcast Control Channel

Before an aircraft enters an LDACS cell it has to receive a beacon message sent by the GS via the BCCH, which enables handling out physical parameters between AS and GS for establishing principle communication. As this is the very first message arriving at the AS, we need to make sure it comes from a valid entity. For this reason we recommend using a combined approach with digital signatures embedded in the broadcast control message

by the GS and the Timed-Efficient Stream Loss-tolerant Authentication (TESLA) protocol [22].

2) Trust

We have to rely on an overall structure that enables us to establish trust among communication entities. As LDACS and AeroMACS are both FCI candidates, with AeroMACS already strongly deployed in China and having an existing Public Key Infrastructure (PKI) in place [4], we could use that for LDACS as well.

3) Entity Authentication

We propose to use the signed Station-to-Station (STS) protocol [2] with preinstalled certificates on the end-entities.

4) Key Negotiation

The dual functionality of the STS protocol is used here.

5) Key Derivation

For generating session keys of arbitrary length with high entropy, we recommend the HMAC-based Extract-and-Expand Key Derivation Function (HKDF) [16].

6) Confidentiality Protection of Messages in Transit

AES-256 in Galois Counter Mode (GCM) is recommended. It also covers integrity and authenticity protection of messages in transit with the same algorithm.

7) Integrity and Authenticity Protection of Messages in Transit

AES-256-GCM is recommended here, as well.

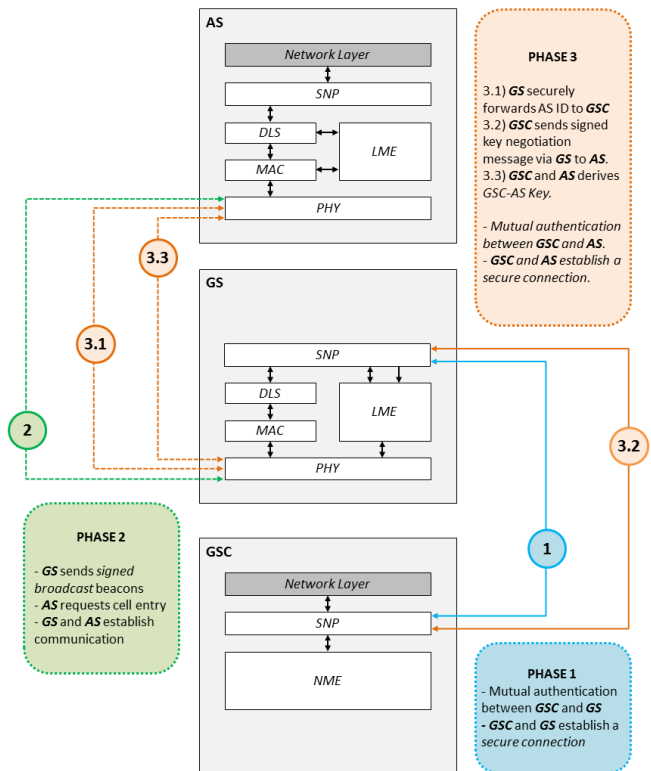


Fig. 6. Steps performed to gain security in LDACS

Figure 6 illustrates which task has to be performed in which order by which entity to establish a secure connection between all relevant LDACS entities. GS and GSC set up a secure connection (Phase 1). Now the GS is identified, mutually authenticated with the GSC and can start broadcasting its signed and TESLA protected beacon to all incoming AS. After that, when an AS approaches a GS, it can verify the authenticity of the messages (TESLA) and the authenticity of the broadcasting GS via its signature (Phase 2). With negotiated parameters for wireless transmission, AS and GS can exchange messages and the AS transmits its identification and authentication information to the GS which verifies them (Phase 3.1). If the verification is successful, it forwards them to the GSC. The GSC also verifies the claim that the AS is actually the entity it claims to be and if successful, starts transmitting the required STS parameters to the AS (Phase 3.2). Now key negotiation, derivation and confirmation can be done between AS and GSC. As soon as this succeeds secure and, thus, encrypted, integer and authenticated communication can commence among authenticated parties (Phase 3.3).

IV. IMPLEMENTATION AND SIMULATION

The security features and the respective message flow described in Section III-A were implemented in a Python event-based software simulation. The tool is capable of simulating world-wide air traffic movements, management of ATC/AOC data, cell coverage, and further features and is called FACTS2 [12]. FACTS2 is a service oriented simulation framework for aeronautical communication system evaluation developed and implemented by DLR. Due to the framework's structure a task separation becomes possible allowing natural parallelization at service level. With this strategy architectural complexity of software design stays low, and integration of existing software tools is fostered. A comparison between data from real flight trials and data from the simulation showed, FACTS2 is able to simulate air traffic movement and aeronautical communication almost as good as in reality and is thus a suitable candidate to test the designed security features for LDACS.

A. Assumptions

For the final implementation of LDACS's cybersecurity architecture in the simulation tool the following assumptions were made:

- (A1) The actual computation time takes no time in the simulation as it is negligible compared to the transmitting time even for the smallest frame in LDACS.
- (A2) LDACS implements an Automatic Repeat Request (ARQ) protocol, thus if frames are lost due to simulated Bit-Error Rate (BER) the data is retransmitted as specified by the LDACS standard for acknowledged send mode.
- (A3) The AS remain in the same range and, thus, the same BER (Bit Error Rate) is used for all scenarios, even though in reality the planes will approach the GS (signal improves, BER is decreasing) and leave the LDACS cell (signal lessens, BER is increasing).
- (A4) The packet generators use exponential distribution to match the requested load on the link, however only transmits dummy data of the actual length of real ATS or AOC data.
- (A5) Certificates for end-entities were already locally stored and are not part of the STS message exchange to reduce the overhead of security data.
- (A6) Before starting with the actual description, it is important to note that in our implementation the most trusted entity is regarded as "server" and the other one as "client". For example if GSC and GS communicate, the GSC is the server and the GS is the client. The resulting order of trustworthiness is then $GSC > GS > AS$.

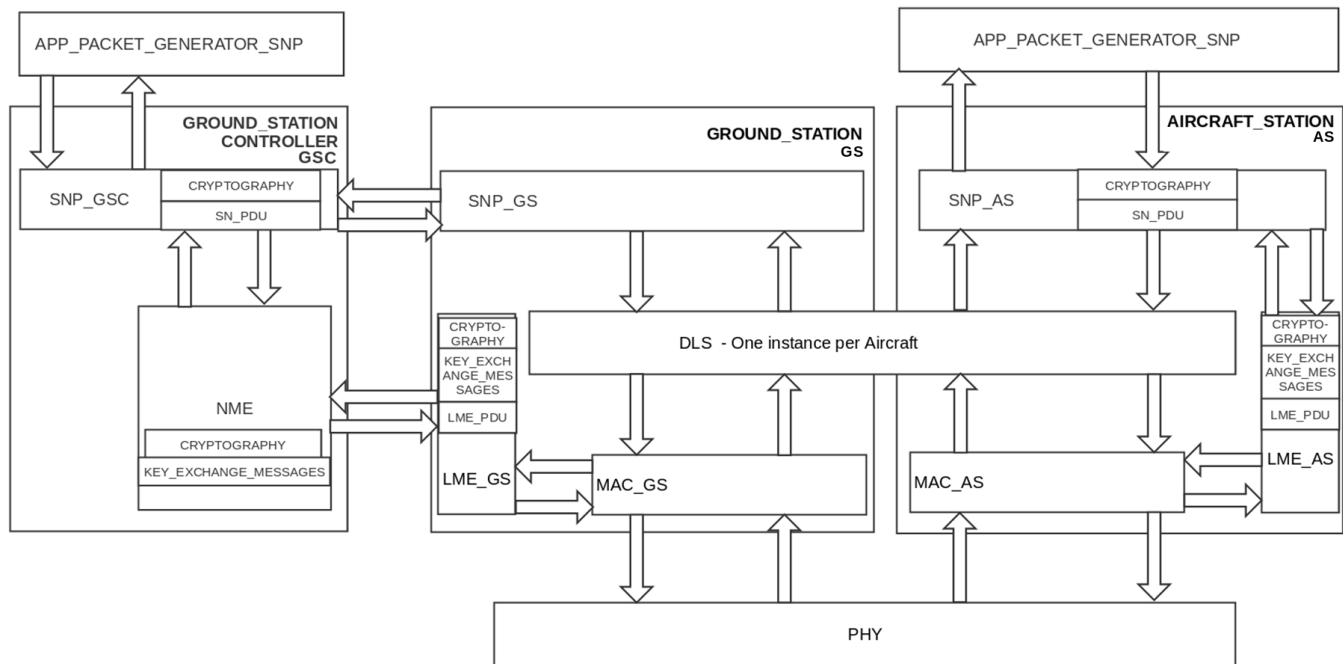
With the assumptions A1-A6 in place the existing simulation tool was extended for the evaluation of the message flow within the LDACS protocol stack. The extension includes the following and results in the data flow shown in Figure 7:

- There is a GSC, one GS, and up to 512 AS connecting to that GS, which in turn is linked to the GSC.
- The message exchange between GSC, GS, and multiple AS can be accurately depicted.
- Data packets are simulated by packet generators (*APP_PACKET_GENERATOR_SNP*), as shown in figure 7, instead being sent from Air Traffic Network Service Providers (ANSP) or airlines in reality.

B. Value Settings

With the adapted protocols and the transition from incoming data packets from the IP Layer (IPv6) to the SNP, we needed to adjust the STS protocol messages for the designed **key negotiation and entity authentication**. We define the required certificates already to be stored upon the LDACS communication devices, so we do not need to include these in the message exchange. The LDACS specification defines certain fields such as "TYPE", "ID", "UA", "PRIO" and so on for SNP and LME Protocol Data Units (PDU), so we adapted those for our security related messages [26]. For naming and structure of our messages we looked at TLS 1.3 [23], giving the more trusted entity the title "server" e.g. the GSC, and the less trusted entity the title "client" e.g. the AS. Therefore, we can start with the SignedClientHello message, which has the structure and parameters selected as listed in Table II. For the ServerHelloKeyExchange message, depicted in Table III, we integrated also parts of the key exchange message to reduce the total amount of exchange packets. Regarding the two KeyExchangeFinished packets (cf. Tables IV and V) we follow standard STS message exchange without changes. These four messages of table II - table V are exchanged between the *NME* of the GSC and the *LMEs* of AS and GS as depicted in Figure 7.

All incoming data from the network will be encapsulated in so called *SN_PDUs*. The packet structure and its parameters is depicted in Table VI and only with applied cryptographic mechanisms exchanged, between the *SNPs* of GSC and AS - as shown in Figure 7.



GSC = Ground Station Controller
GS = Ground Station
AS = Aircraft Station

SNP = Sub-Network Protocol
NME = Network Management Entity
LME = LDACS Management Entity
DLS = Data Link Service
MAC = Medium Access Layer
PHY = Physical Layer

PDU = Protocol Data Unit
SN_PDU = Sub-Network PDU

Fig. 7. Software implementation of the LDACS protocol with security additions

TABLE II
SIGNEDCLIENTHELLO MESSAGE

Field	Size	Description
TYPE	4 bit	SignedClientHello
ID	12 bit	Packet Identifier
UA	28 bit	Unique Address - Packet Sender Identifier
PRIO	4 bit	Priority of packet
RNC	256 bit	Random Number Client
SIGN	128 bit	Signature
GS_VAL	1 bit	Field containing information of GS verification status of AS.

TABLE IV
CLIENTKEYEXCHANGEFINISHED MESSAGE

Field	Size	Description
TYPE	4 bit	ClientKeyExchangeFinished
ID	12 bit	Packet Identifier
UA	28 bit	Unique Address - Packet Sender Identifier
PRIO	4 bit	Priority of packet
GY	2048 bit	g^y , y chosen in secret by sender
ENCSIGGYGX	256 bit	Signature of g^y , g^x encrypted with common key derived from message $key = (g^x)^y \text{ mod } p$ exchange.

TABLE III
SERVERHELLOKEYEXCHANGE MESSAGE

Field	Size	Description
TYPE	4 bit	ServerHelloKeyExchange
ID	12 bit	Packet Identifier
UA	28 bit	Unique Address - Packet Sender Identifier
PRIO	4 bit	Priority of packet
RNS	256 bit	Random Number Server
G	2048 bit	Generator for cyclic group mod p
P	224 bit	Prime Number for cyclic group
GX	2048 bit	g^x , x chosen in secret by sender

TABLE V
SERVERKEYEXCHANGEFINISHED MESSAGE

Field	Size	Description
TYPE	4 bit	ServerKeyExchangeFinished
ID	12 bit	Packet Identifier
UA	28 bit	Unique Address - Packet Sender Identifier
PRIO	4 bit	Priority of packet
ENCSIGGXGY	256 bit	Signature of g^x , g^y encrypted with common key derived from message $key = (g^y)^x \text{ mod } p$ exchange.

In the implementation we work with so called **signed beacons**. Hence, for securing parts of the data channels,

we suggest putting a signature in the broadcast slots in the Broadcast Channel BCCH), used together with the TESLA protocol. This allows incoming aircraft on the first receipt

TABLE VI
SN_PDU MESSAGE

Field	Size	Description
TYPE	4 bit	ACK/UNACK/MGMT
PID	5 bit	Packet ID
SC	3 bit	Service Class
SAC	28 bit	Subscriber Access Code
LEN	12 bit	Octets PDU
USER DATA	844-12108 bit	Payload (optionally encrypted)
MAC	128 bit	Integrity Checksum

of a GS announcement the verification of the identity of the broadcasting GS. The corresponding packet structure is shown in Table VII, with Figure 7 showing, this broadcast message can only be sent from the *LME_GS* and is received by the *LME_AS*.

TABLE VII
TESLA PROTECTED SIGNED SYSTEM IDENTIFICATION BROADCAST MESSAGE

Field	Size	Description
B_TYPE	4 bit	System Identification Broadcast
LEN	10 bit	Length given in bit
GS SAC	12 bit	GS Identifier - Identifier who sent the packet
VER	3 bit	Protocol Version
FLF	12 bit	Forward Link Frequency
RLF	12 bit	Reverse Link Frequency
MOD	1 bit	User-/Cell-specific ACM
CMS	3 bit	Coding and Modulation Scheme
EIRP	7 bit	GS Equivalent Isotropic Radiated Power
PAD	0 bit	Reserved
CRC-8	8 bit	Cyclic Redundancy Check
SIGN	128 bit	GS Specific Signature
MAC	128 bit	TESLA authenticated packet proof
TESLA_KEY	128 bit	TESLA key for respective interval

With the described messages in place, a message exchange can be performed enabling secure key negotiation, entity authentication and finally the confidential, integrity protected transmittance of user data.

V. EVALUATION

First evaluation of data showed promising results regarding the following points which are part of the objectives mentioned in the official LDACS SARPS (Standards and Recommended Practices) by ICAO [17]:

- 1) For time overhead added by the entity authentication, key agreement, negotiation and derivation, we wanted to show that it takes less than ten seconds to securely connect an AS to the LDACS network. This requirement emerges from the $RCTP_{CSP} = 10s$ for RCP130/A1 typed messages, originating from DO-350A [25].
- 2) Regarding data overhead by additional security we wanted to use less than five percent of additional user-data capacity. This requirement originated from the SESAR Security Risk Assessment of LDACS performed in April 2019.

A. Time Overhead by Security Additions

Here we wanted to measure the the time until an aircraft is authenticated by and to all other entities and until AS and GSC have a key. We tested this with a total number of 512 AS, filling up an entire LDACS cell, allowing us to measure different connection and authentication times due to load.

Further we checked four different Bit Error Rates (BER), to measure the impact of the BER - and in the end the quality of the channel between AS and GS - on secure communication establishment performance. BER steps were chosen according to previous work [13], ranging from 0, 10^{-6} , 10^{-5} and 10^{-4} . However the BER should only start to be noticeable around a BER of 10^{-5} . LDACS is designed for a BER up to 10^{-4} to work and on average, a BER of 10^{-6} is assumed [12]. With 512 ASs in place we observed that the FL-/RL-Load, the size of SN_PDU and the acknowledged transmission were either constant or irrelevant values and, thus, we decided to altered only the following parameters for additional trials:

- Simulation runs = 10 times
- Simulation time = 1000 seconds
- BER varies between the values 0, 10^{-6} , 10^{-5} , and 10^{-4}

Finally, we ended up with four simulation scenarios. Each one was performed ten times in order to receive reproducible results. We investigated how long the time spans are for (i) GS and AS connection, (ii) GSC and AS connection, (iii) GSC AS-GSC master key derivation, and (iv) AS AS-GSC master key derivation and key confirmation, in respect to the different BER. After those four steps were successfully performed key derivation takes place and a secure session can be established, which takes zero processing time in our model, thus we neglect the derivation times.

As the received measurements illustrated on logarithmic scales in Figures 8a and 8b prove, a BER of 0 or 10^{-6} has little effect on the overall connection time for establishing a secure link. With an increasingly bad channel and thus a higher BER of 10^{-5} , Figure 8c shows that on average the overall connection time remains the same as with lower BER, however the longest connection times spike up and take already 50% more time than with 10^{-6} BER. In Figure 8d, we can see that LDACS was designed to operate up to a maximum of one bit error per 10,000 bit and, thus, the very high connection and security exchange times due to retransmissions and waiting times with a BER of 10^{-4} are not surprising. The most important detail here is, that a secure LDACS link can still be established under such bad channel conditions.

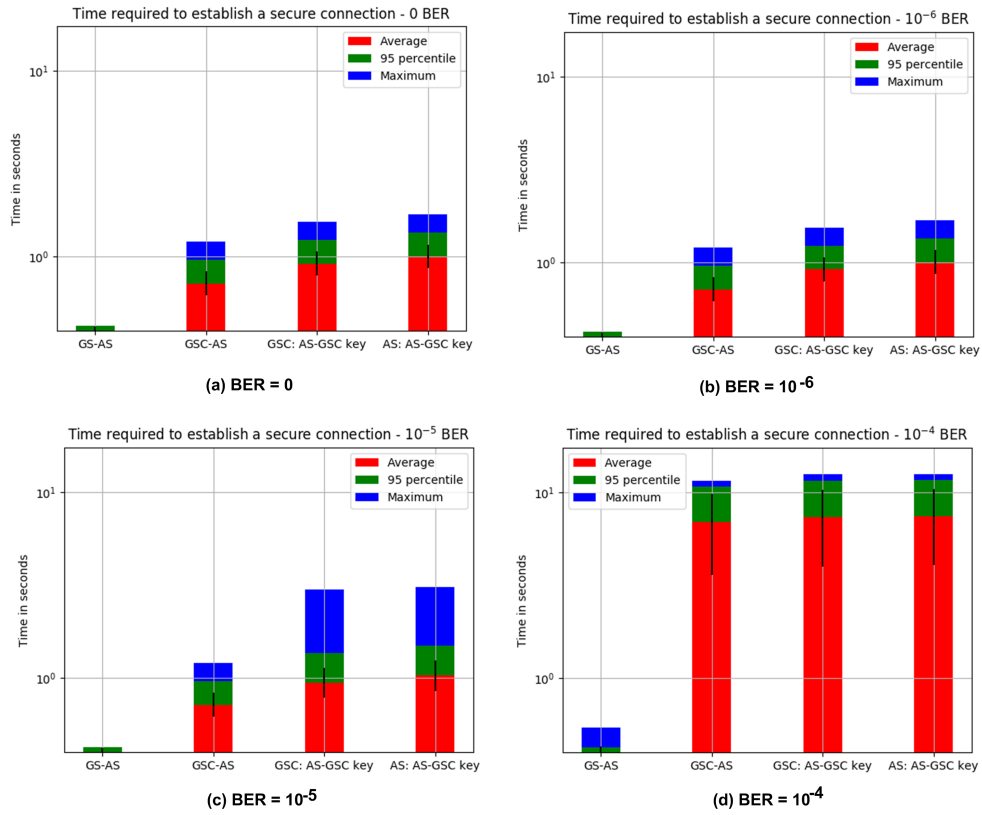


Fig. 8. Duration of AS, GS and GSC mutual authentication, key negotiation and secure connection establishment time

B. User Data Overhead by Security Additions

Security data overhead is introduced by having additional authentication and key negotiation packets in the system and by using Message Authentication Codes (MAC) of 128 bit attached to data packets. Also additionally required re-transmission depending on the BER due to larger or more packets due to security data overhead can have an impact on security data overhead. We use the following fixed parameters for our simulation. The choice on 100 AS and 100 kbit/s on FL and RL emerged from previous work [12].

- 100 AS
- 100 kbit/s for average FL/RL data throughput for all AS in one LDACS cell
- 10 simulation runs
- 500 seconds simulation time
- acknowledged transmission mode
- BER = 10^{-5} to simulate ten times worse conditions than the average assumed BER
- 128 bit MAC size

We alter only the size of the SN_PDUs between 128 byte, 256 byte, 384 byte, 512 byte, 640 byte, 768 byte, 896 byte, 1024 byte, 1152 byte, 1280 byte, 1408 byte, and 1536 byte. For each mutual authentication and key negotiation, 964 byte must be transmitted between AS and GSC via the GS. Assuming on average, every 500 seconds a new key

would have to be negotiated, as the AS leaves one GS range and enters another, we receive the security data overhead in dependence of the SN_PDU size.

In Figure 9 we see that depending on the size of SN_PDU packets, the security overhead can range from 13.2% to 1%. This mostly results from the large fixed size of MACs attached to user data packets.

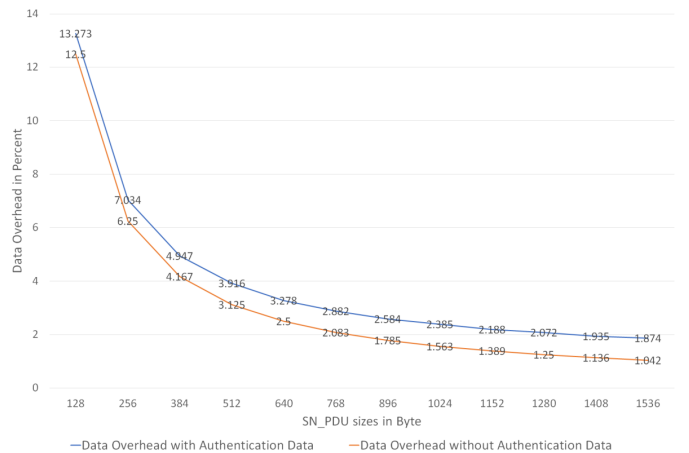


Fig. 9. Security data overhead on top of user data

To achieve our goal of less than 5% security data overhead, we recommend only using user data packets with a size larger than 256 bytes. This is possible as the SN_PDU size is chosen by LDACS subroutines and depends on the perceived BER - the better the BER, the larger the SN_PDUs and vice versa. Thus our results indicate a recommendation of a minimum SN_PDU size for worst-case BER of 384 byte - or to lower the overall MAC size. Also we see that the average re-authentication time for moving nodes such as aircraft of 500 seconds only puts another 0.8% more security data on the link.

Overall with this first evaluation of the implementation of the LDACS cybersecurity, we conclude that reducing amounts of authentication times, security related message exchanges and the sizes of MACs will be critical to further reduce the security overhead on LDACS.

VI. SUMMARY AND CONCLUSION

In this paper we stated that security support is essential for future air traffic and related ATM. Therefore, we introduced a cybersecurity architecture for LDACS offering basic security functionalities (e.g., protection of control channels, key negotiation and derivation) as described throughout Section III. The designed and implemented security solution was successfully tested within the FACTS2 simulation framework. With the results from several simulation runs, we could prove that in the current implementation, the introduced security data overhead by key exchange messages and Message Authentication Codes (MAC) applied to data packets, is nearly negligible. We beat the ten second mark for time overhead with the 95% percentile up until a BER of 10^{-5} and the five percent mark for data overhead with SN_PDU sizes larger than 256 bytes. At a realistic Bit Error Rate (BER) of 10^{-5} , the average secure connection establishment time is 1.05 seconds. Assuming an internal data packet size of SN_PDUs larger than exactly 320 byte, we also beat the five percent margin, making the proposed security architecture viable.

As this work also presented a first proof-of-concept for the viability for the LDACS cybersecurity architecture, it will proof to be valuable input for the SESAR wave 2 specification of LDACS prior to its finalization and deployment as the terrestrial datalink for civil aviation.

During the MICONAV flight campaign this year in March 2019, flying LDACS the first time worldwide, additional data (e.g., evaluation on broadcast security of LDACS, use of Post-Quantum Cryptography (PQC) in digital aeronautical datalinks) was collected that will be further investigated and evaluated. In order to improve LDACS's cybersecurity architecture further, we will have a closer look at post-quantum cryptography, a standalone trust solution for LDACS and additional protection for control messages and physical layer protection against jamming, spoofing and interfering with the datalink.

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